

HealthConnect Week 4 Project Summary

AnalystLab Africa Experience Lab — Week 4 | Data Analytics Track

1. Specific Problem Addressed

The Data Analytics track will investigate appointment attendance and no-show patterns at HealthConnect Clinic, using appointment, booking, reminder, historical and operational variables to identify patterns that can support later decisions aimed at reducing missed appointments and improving appointment utilisation.

2. Resources Used

- AnalystLab Africa Week 4 Experience Lab Assignment.
- HealthConnect_Appointment_Data.csv.
- Official HealthConnect Data Dictionary.

3. Key Week 4 Observations

- The dataset contains 5,000 appointment records, 18 variables and 1,696 unique anonymised patients.
- appointment_id functions as the primary key with no duplicate IDs detected.
- patient_id may recur across multiple appointments.
- appointment_outcome contains Attended, No-Show and Cancelled.
- Initial outcome counts are 2,314 Attended, 2,423 No-Show and 263 Cancelled.
- reminder_channel None is an expected state when no reminder was sent.
- booking_lead_days is internally consistent with booking_date and appointment_date.
- previous_no_shows does not exceed previous_appointments, matching the official constraint.
- Distance to clinic and waiting time contain limited missing values.

4. Proposed Analytical Approach

1. Use the Data Dictionary as the authoritative variable-definition reference.
2. Preserve the source CSV and create separate processed data for transformations.
3. Validate the analysis-ready data before calculating measures.
4. Keep Attended, No-Show and Cancelled distinct.
5. Calculate the selected KPIs in the next analysis stage.
6. Investigate no-show patterns by appointment type, lead time, previous no-shows, reminders/channel, day, time, age group, gender and distance.
7. Assess waiting time as an operational variable.
8. Interpret descriptive relationships as associations unless stronger evidence is available.

5. Key Considerations

- The data are synthetic/anonymised.
- Repeated patient IDs require distinction between patient-level and appointment-level analysis.
- Missing distance and waiting-time values require documented treatment.

- Reminder-channel None is a defined state.
- Cancelled and No-Show outcomes should remain distinct.
- Simple comparisons cannot establish causality.

6. Proposed Week 5 Focus

Move into validated analytical work: prepare the analysis-ready dataset, calculate the agreed KPIs, examine no-show patterns across priority dimensions, identify actionable findings and develop outputs according to the official Week 5 instructions.